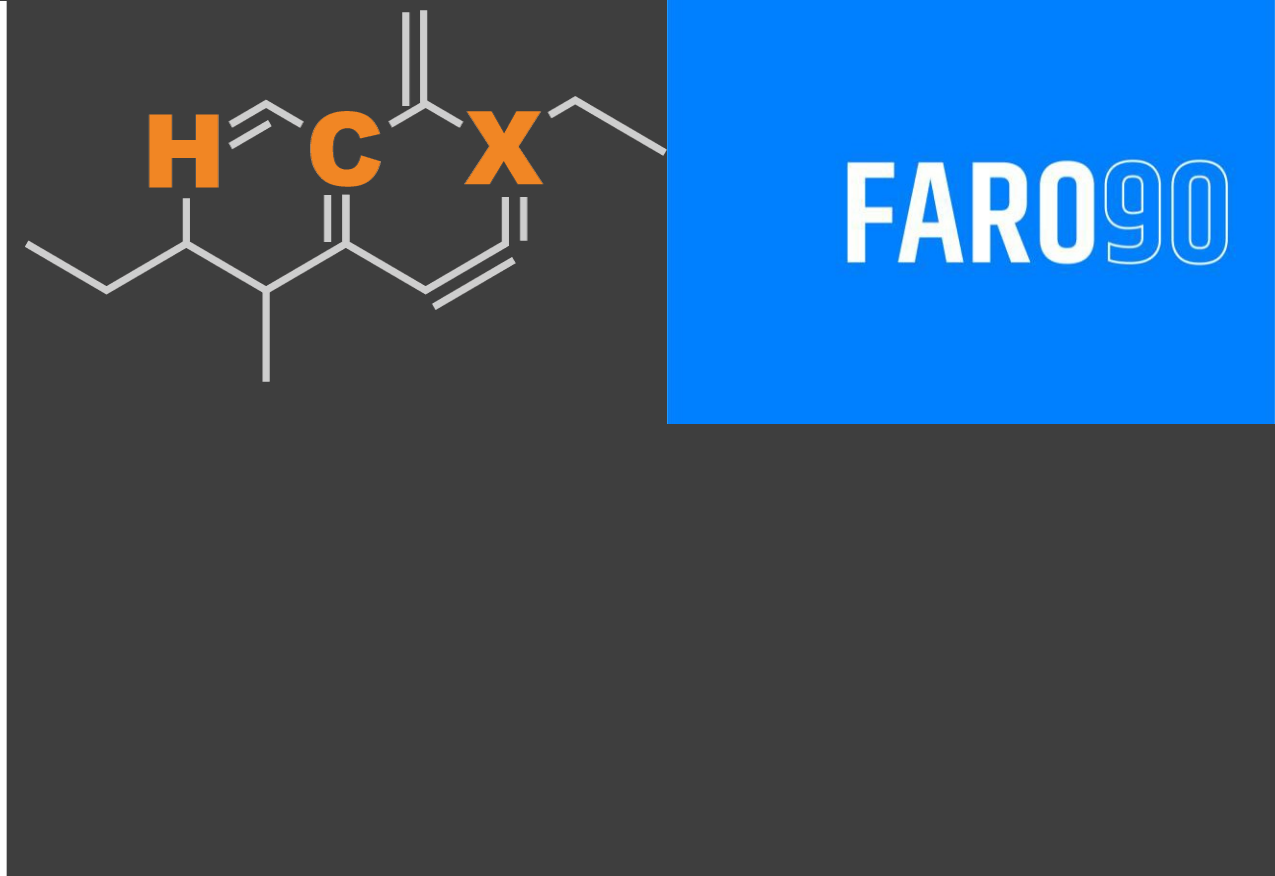




Ethanol Blending in Gasoline - Latvia

September, 2025

Ethanol Blending in Europe

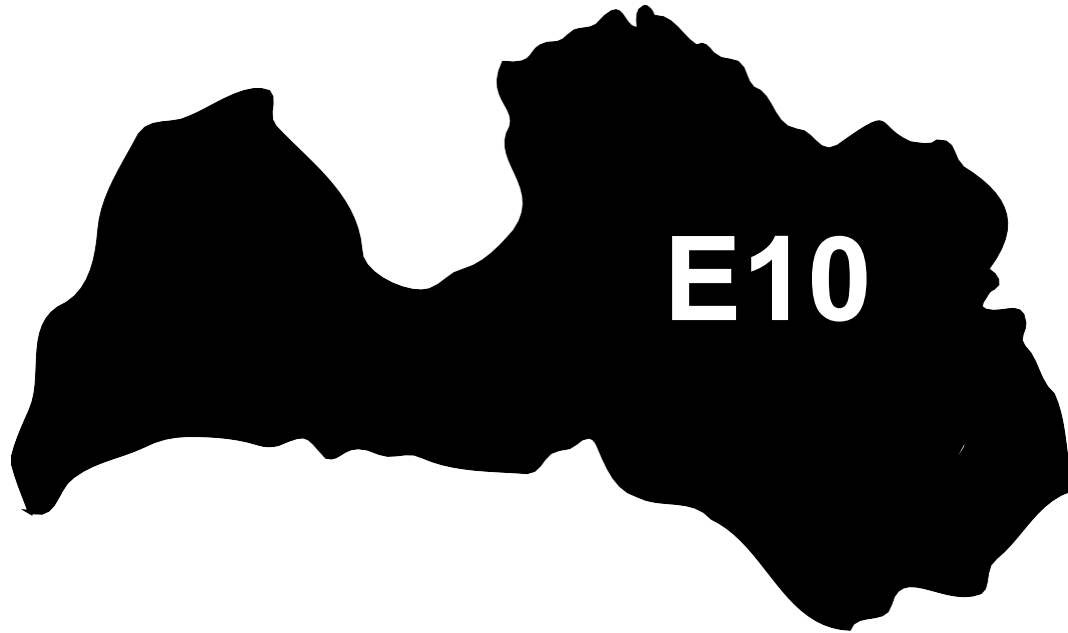
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Latvia

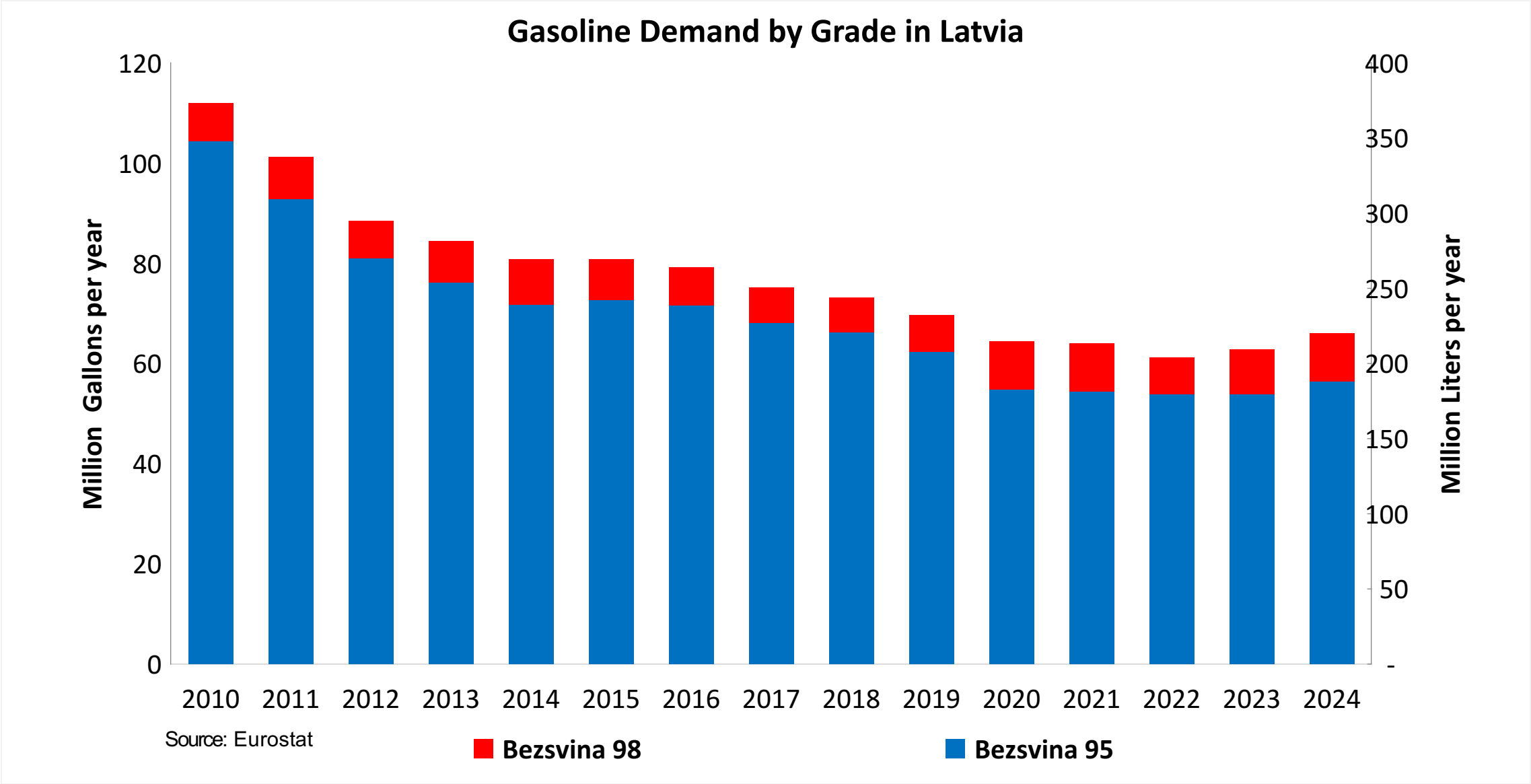


In 2024, gasoline consumption in Latvia was 220 million liters (58 million gallons) and growth rate was -1.1%. There is no gasoline production. Gasoline net imports were 200 million liters (53 million gallons). Latvia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.4 RON.

In 2024, ethanol consumption in Latvia was 23 million liters (6 million gallons) representing 10.4% of gasoline demand. Ethanol production and Net Imports were 25 and -3 million liters (6 and -1 million gallons) correspondingly. Ethanol source Cereals 100%.

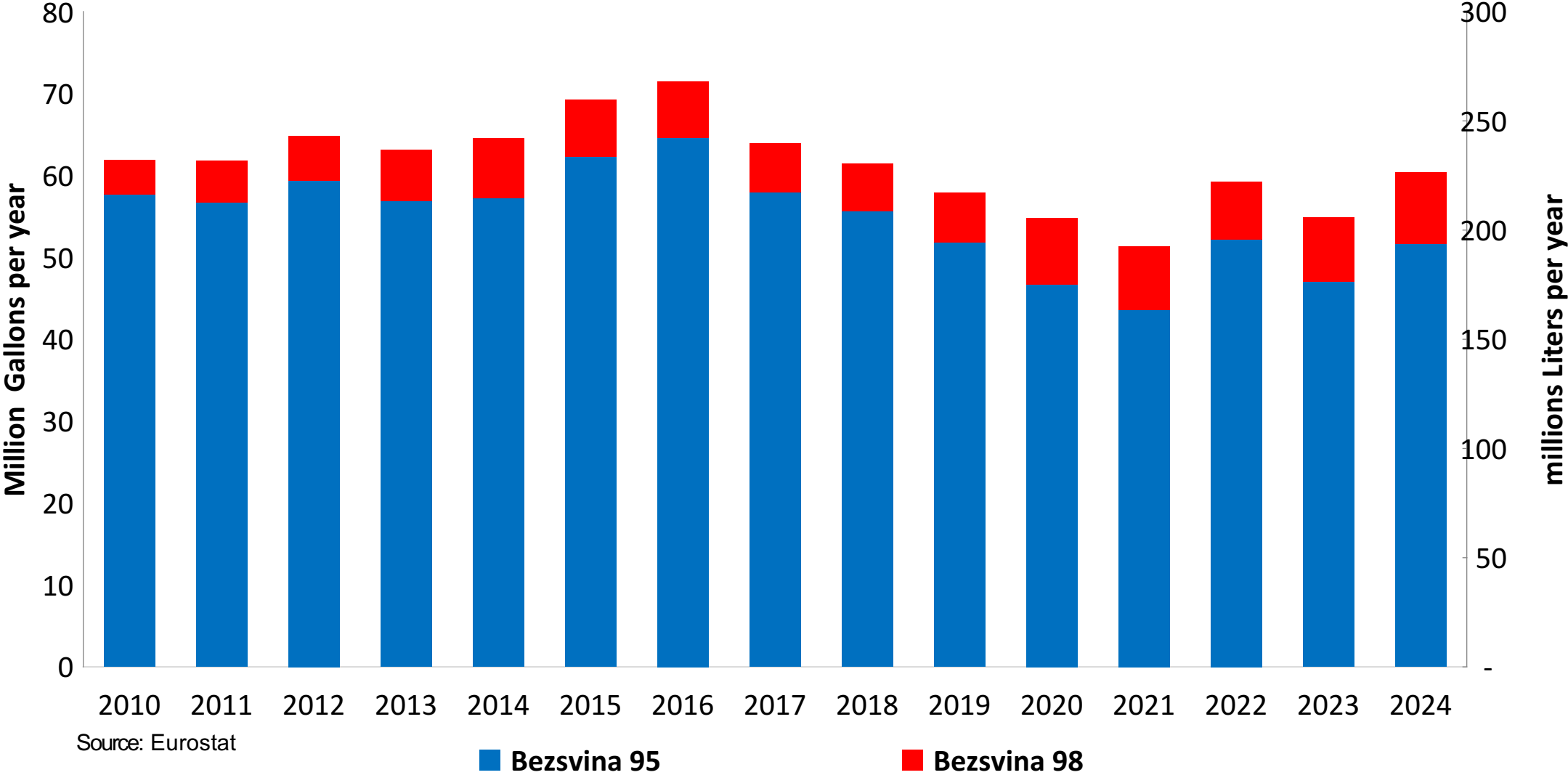
Source: Eurostat, European Commission

Gasoline Demand in Latvia



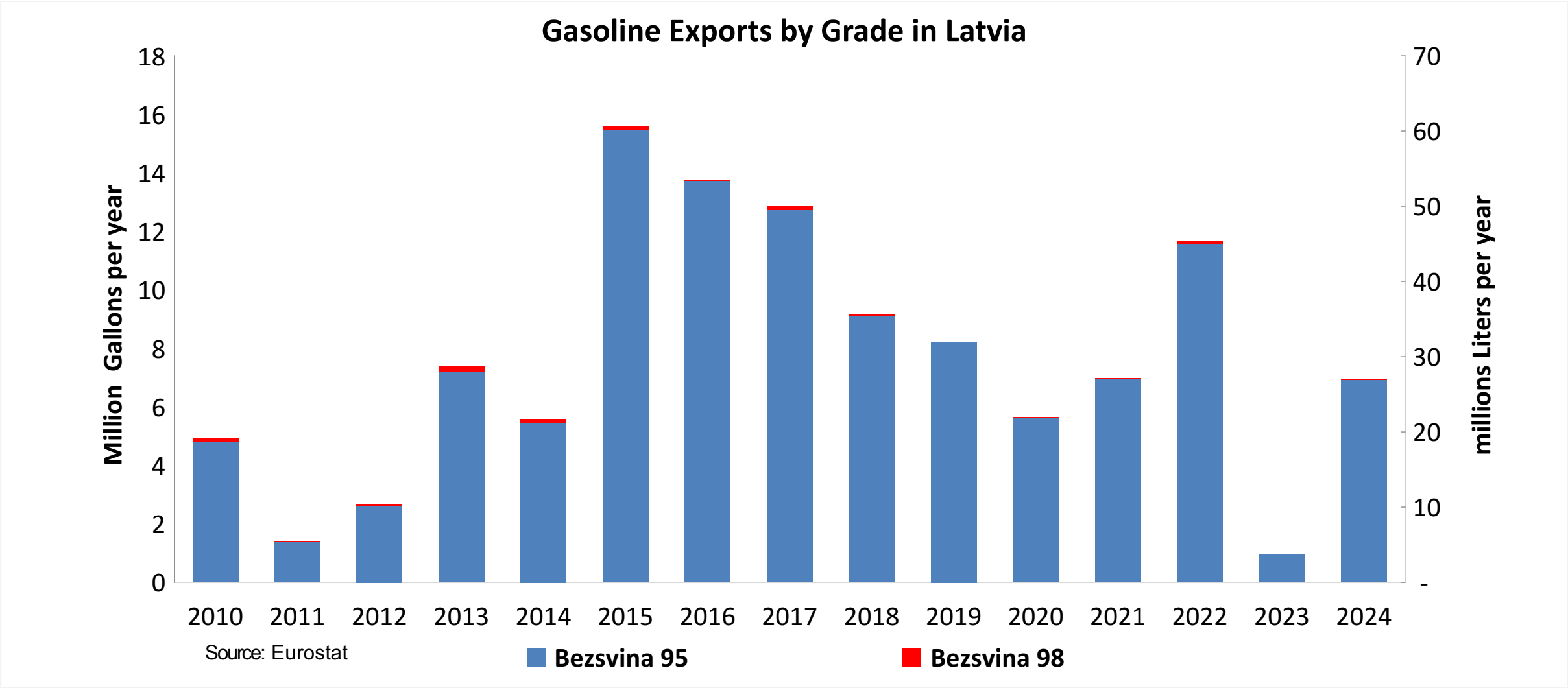
Gasoline Imports in Latvia

Gasoline imports by Grade in Latvia

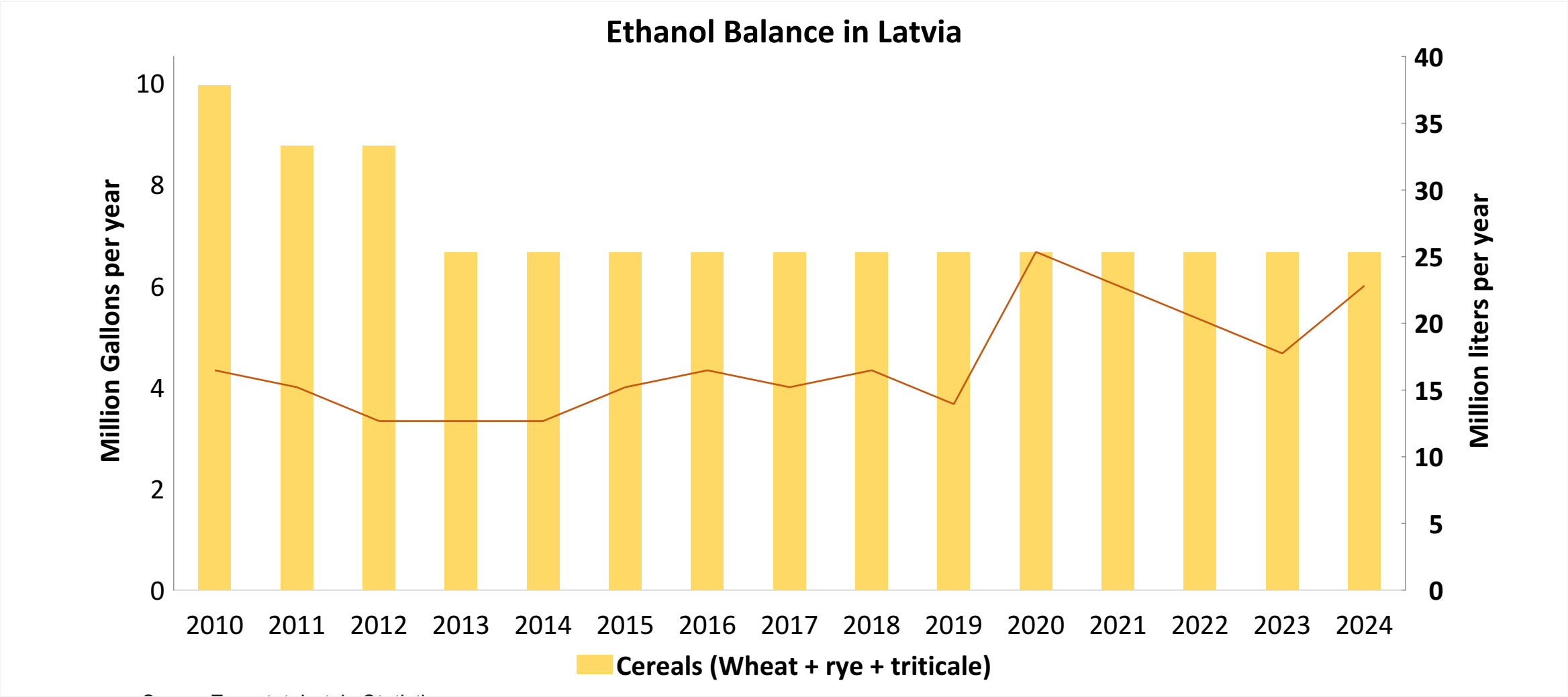


Source: Eurostat

Gasoline Exports in Latvia



Ethanol Balance in Latvia



Source: Eurostat, Latvia Statistics

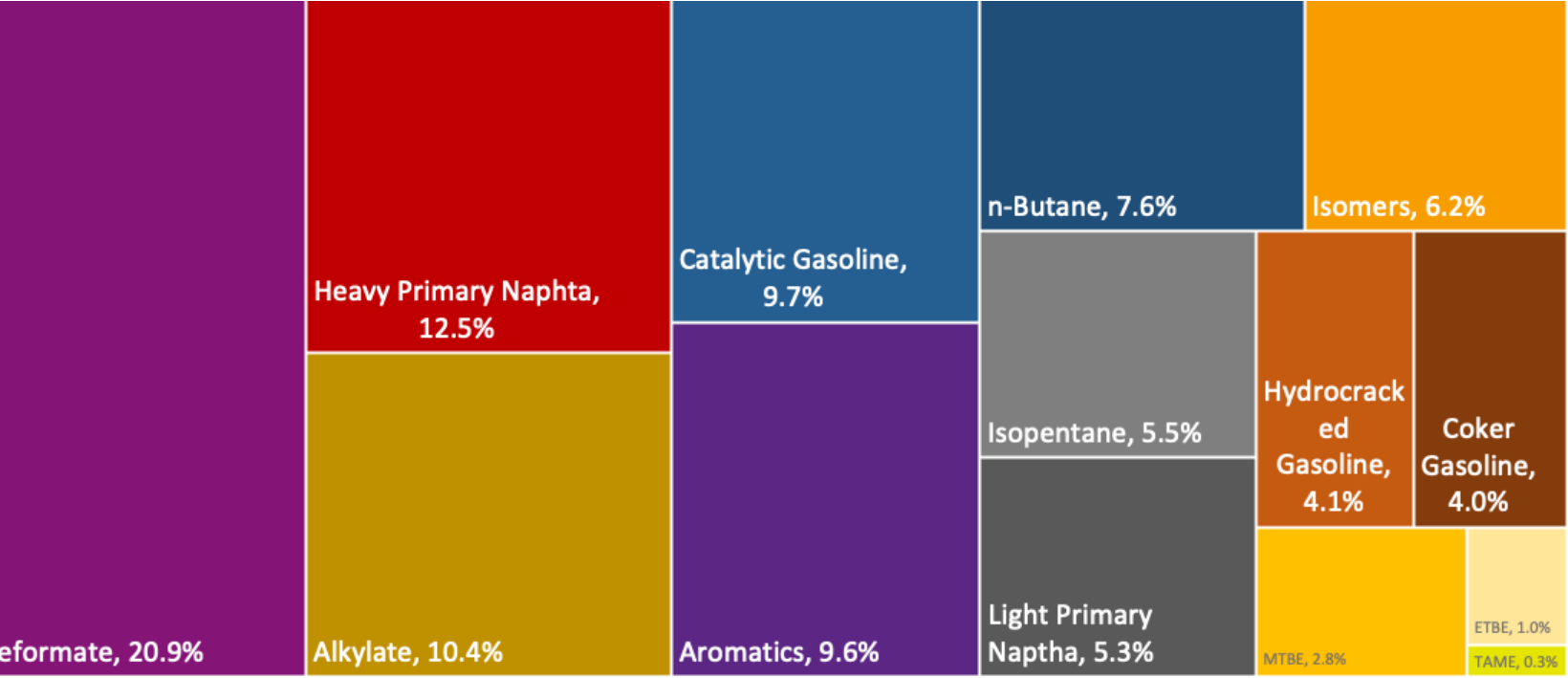
Gasoline Quality in Latvia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

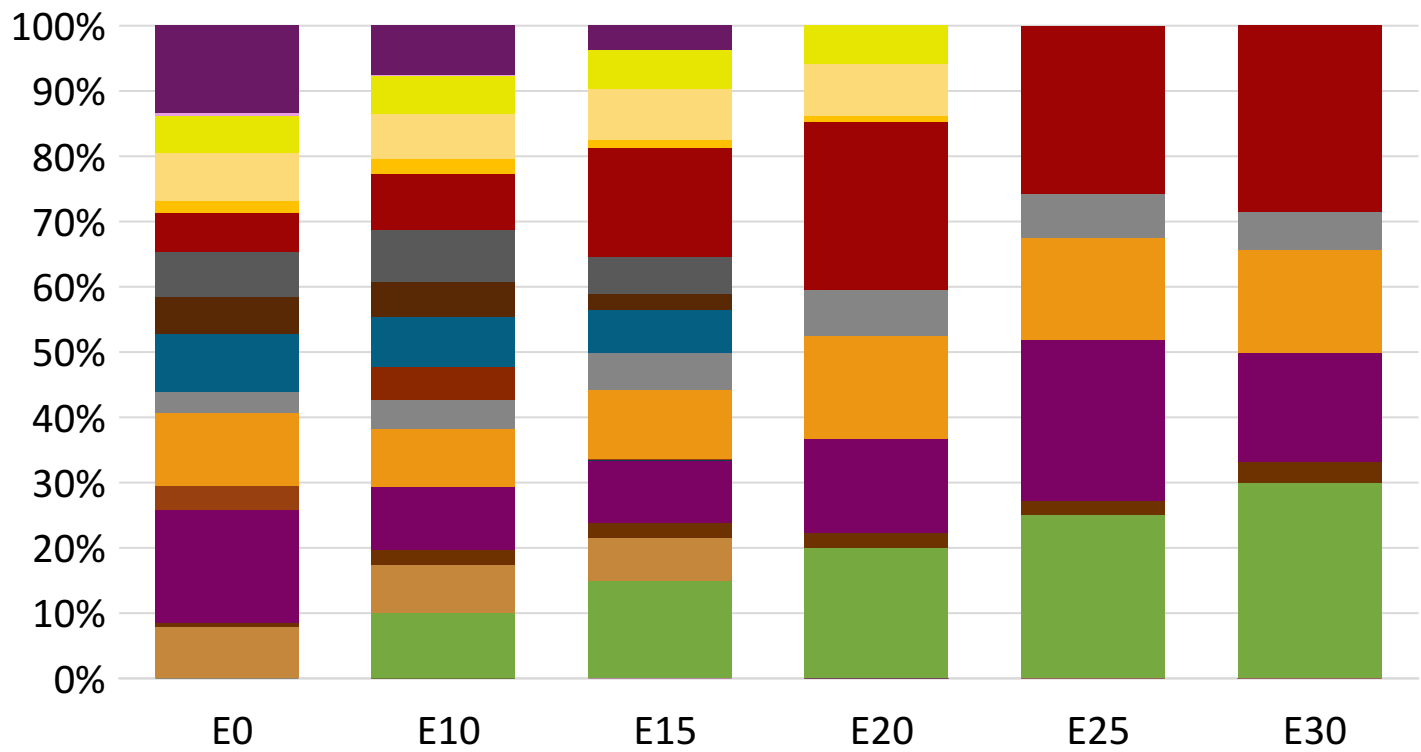
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Latvia Available Blending Components



Latvia – Gasoline 95 – Constant Octane

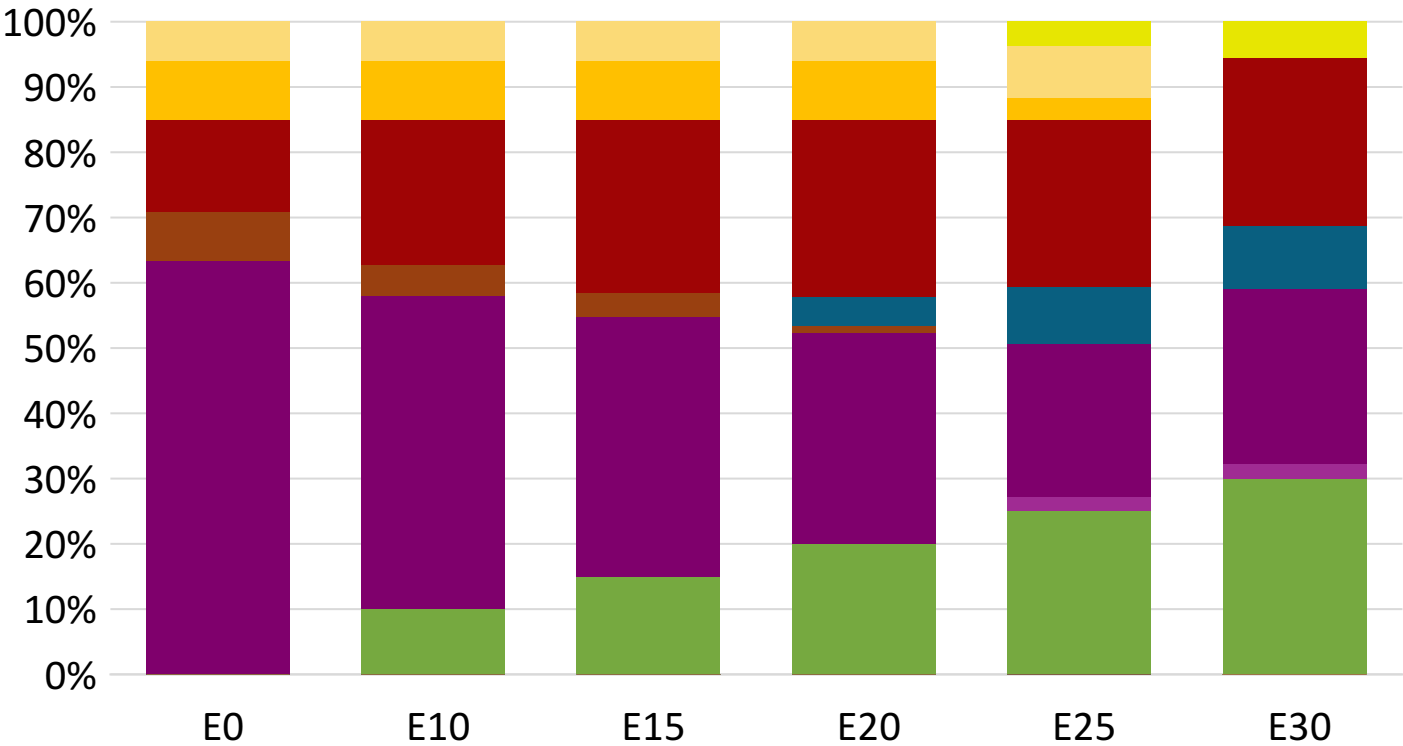
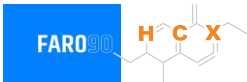


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Latvia - Gasoline 98 - Constant Octane

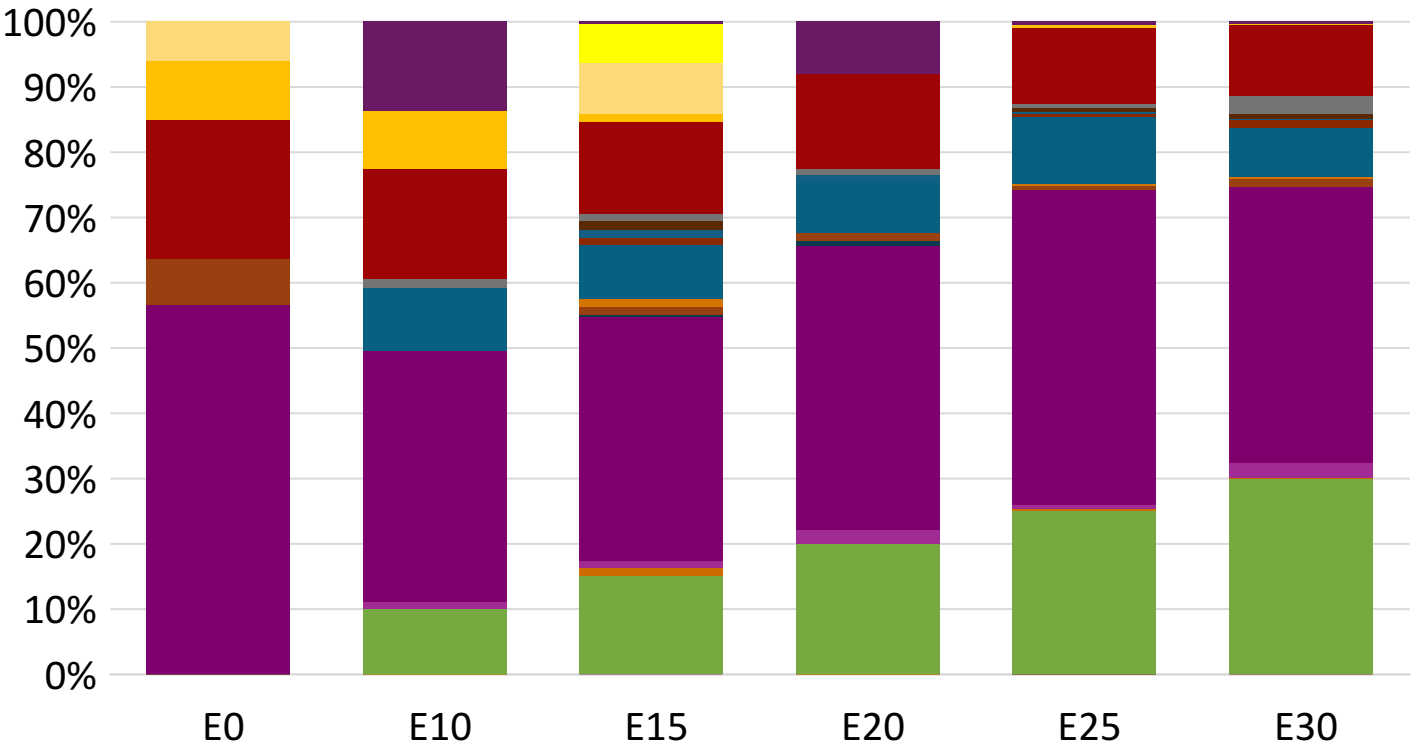
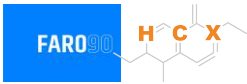


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Latvia – Gasoline 95 – Increased Octane

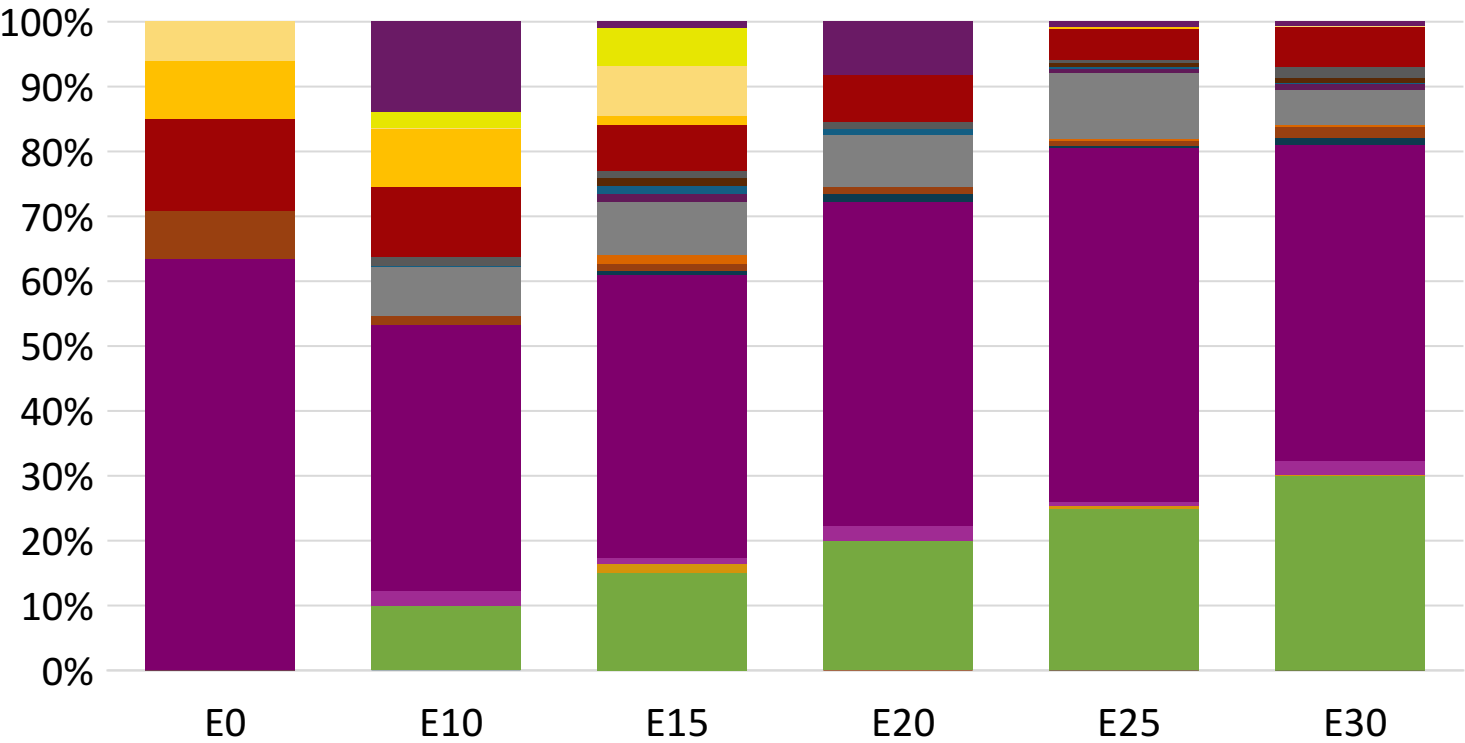


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Latvia – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

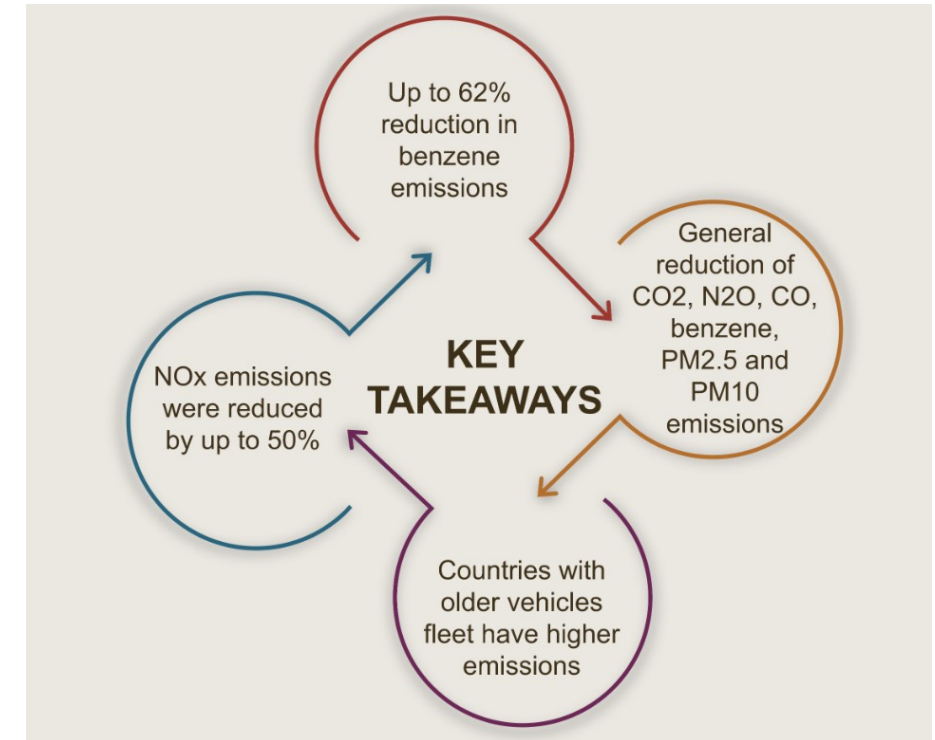
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

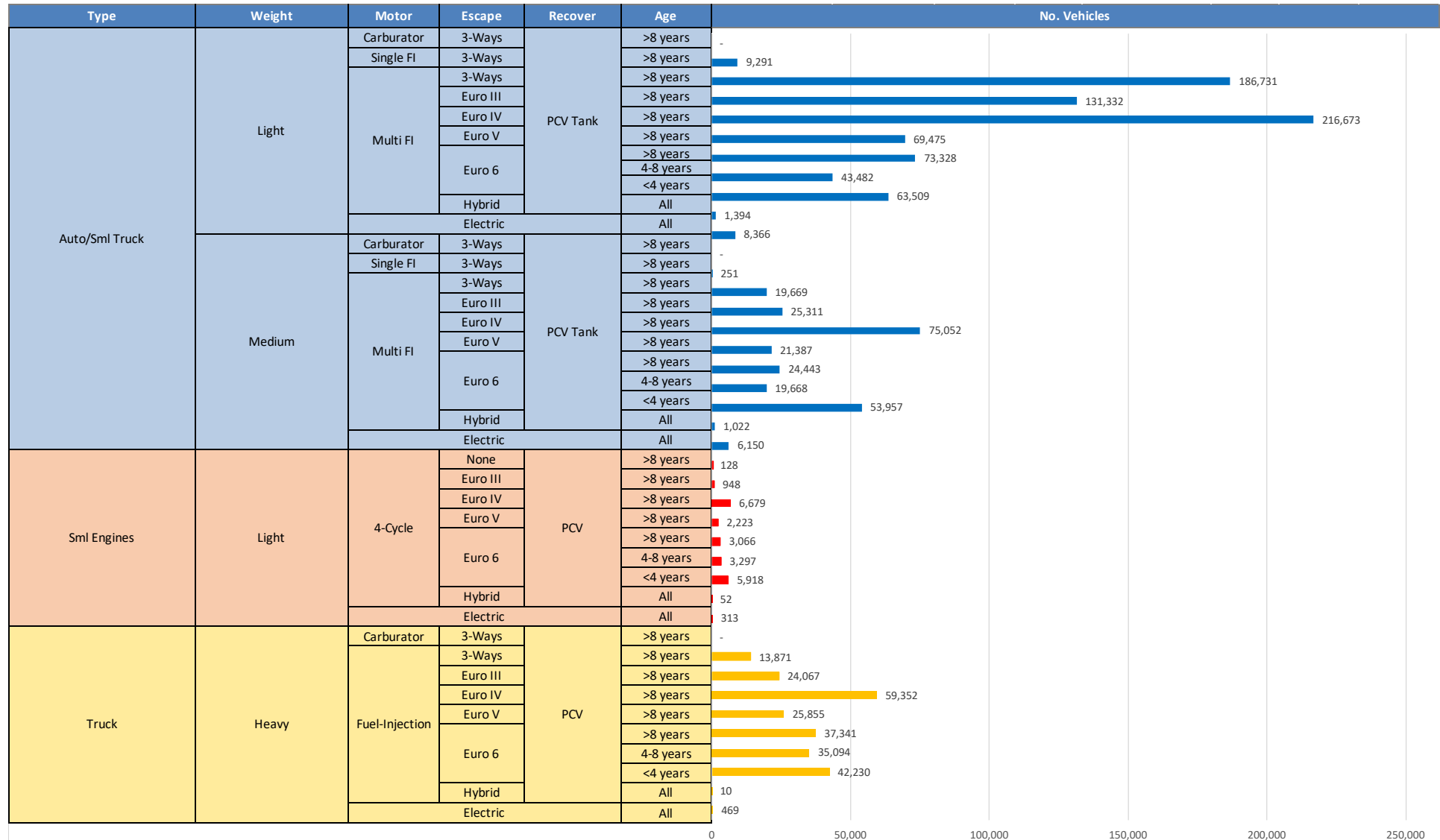
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Latvia

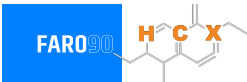


Vehicle Fleet: 1,311,403
Gasoline Vehicle Fleet: 333,918

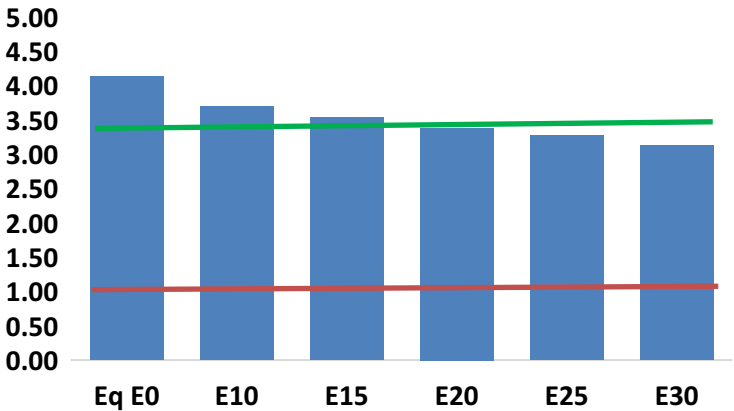
Source: Eurostat, ACEA

Latvia – Vehicular Emissions

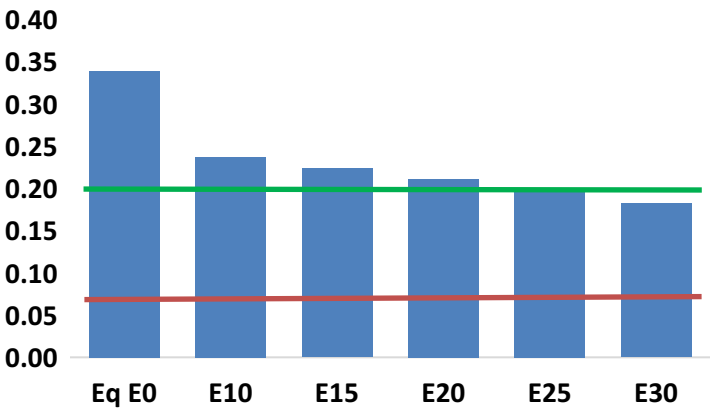
TIER III BIN 125 United States
Euro 6 Standard



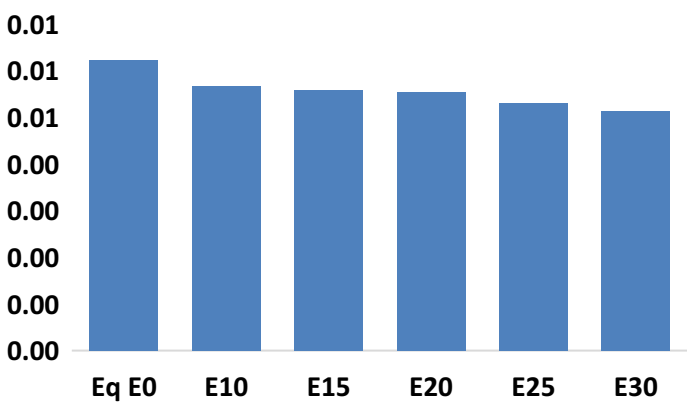
Carbon Monoxide (CO) g/km



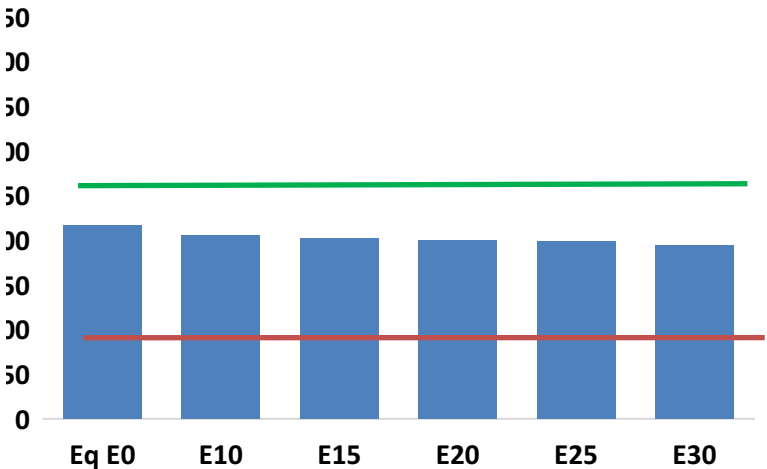
Nitrogen Oxides (NOx) g/km



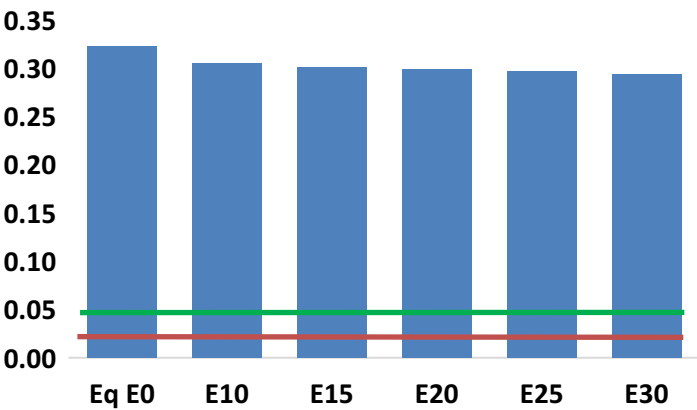
Aromatics (BTX) g/km



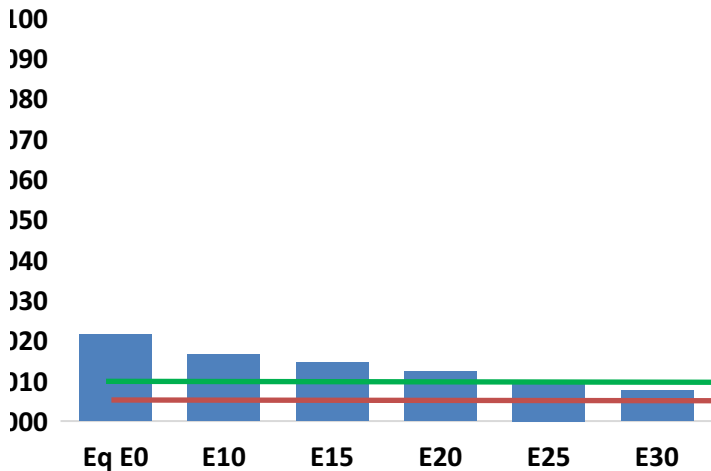
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



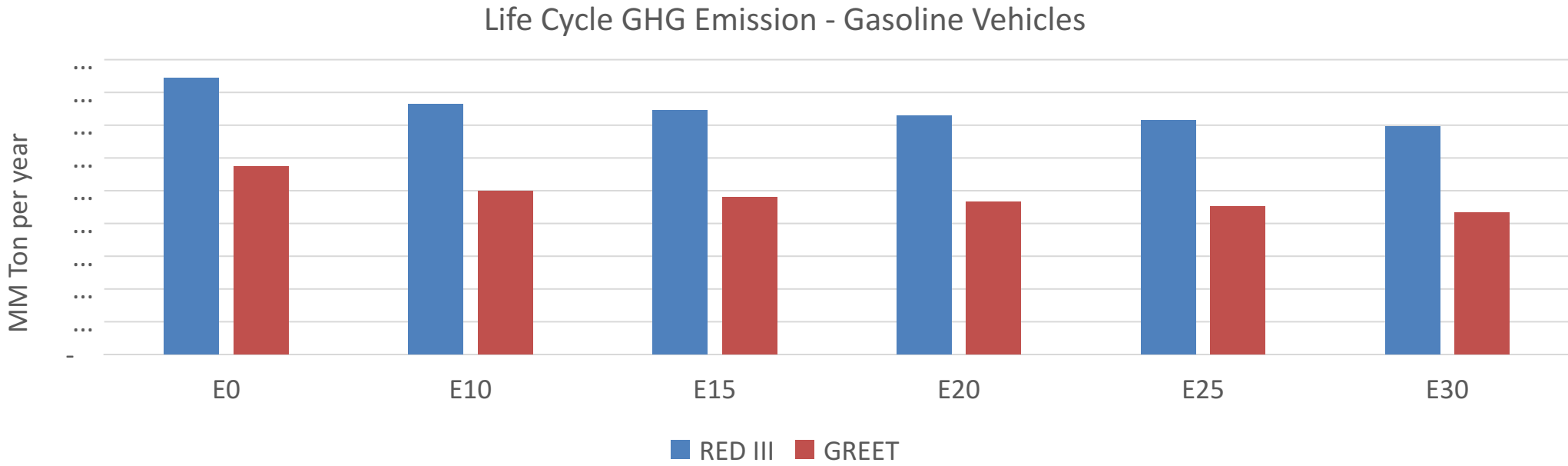
Particulate Matter (PM2.5) g/km



Latvia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	36,253	34,335	32,417	31,014	29,681	28,680	27,419	-11%	-18%
CO2	1,902,185	1,853,272	1,807,076	1,770,744	1,752,766	1,741,942	1,709,833	-5%	-8%
VOC	2,835	2,759	2,682	2,646	2,624	2,610	2,578	-5%	-7%
NOx	2,975	2,232	2,083	1,963	1,851	1,728	1,598	-30%	-38%
SOx	23	21	19	18	16	15	13	-15%	-28%
NH3	639	633	626	629	636	641	646	-2%	0%
N2O	29	29	29	29	30	30	30	-1%	2%
CH4	629	629	629	638	652	664	676	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	35	34	34	33	33	33	33	-4%	-4%
Acetaldehyde	68	78	114	174	237	270	319	68%	249%
Formaldehyde	197	191	223	262	274	303	330	13%	39%
Benzene	55	53	50	49	49	46	45	-9%	-11%
PM2.5	117	104	91	79	67	54	41	-22%	-43%
PM10	71	63	55	48	41	33	25	-22%	-43%

Latvia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	12.1
Target @ 2035 (MMT/year)	6.8
Delta	5.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.1%	16.2%	18.7%	21.1%	24.3%
RED II/III	0.0%	9.3%	11.6%	13.4%	15.2%	17.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.5%	4.0%	4.6%	5.3%
RED II/III	0.0%	3.0%	3.7%	4.3%	4.8%	5.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90